

University of Transport Ho Chi Minh City



Deep Learning

Course Report

Detecting Deepfakes with Vision Transformers and Convolutional Neural Networks



Lecturer: PhD Nguyen Thi Khanh Tien

Students

067206006852	Nhữ Phạm Quang Mạnh
056206009261	Trương Đỗ Vương
079204034414	Lê Hoàng Tuấn
082306000670	Nguyễn Ngọc Phương Uyên
064206001079	Nguyễn Trần Việt Nhật
054206006377	Nguyễn Đình Khánh Nguyên

Ho Chi Minh City, 2026

Abstract

Deepfake detection is a challenging binary classification problem in which a model must distinguish authentic face images from synthetically manipulated ones. This coursework implements and compares two families of vision backbones—a Vision Transformer (ViT) and a modern Convolutional Neural Network (ConvNeXt)—for the task of facial manipulation detection. Both backbones are initialized from the self-supervised DINOv3 weights released by Meta, are distilled from the same ViT-7B teacher, and are evaluated at a matched parameter budget of approximately 28M parameters, so that any observed performance difference is attributable to the inductive bias of the architecture rather than to scale or training signal. We evaluate the frozen backbones with a linear probe trained on the DeepFakeFace dataset and report cross-identity accuracy, precision, recall, F1 and ROC-AUC. Our results show that at a matched parameter count the two architectures perform nearly identically on a diverse, modern manipulation benchmark (74.1% vs. 74.5% F1), while the convolutional backbone substantially outperforms the transformer on an older faceswap-GAN dataset (98.9% vs. 75.8% detection rate), consistent with the hypothesis that convolution is better suited to capturing the local, low-level blending artifacts that dominate earlier generation deepfakes. We further show that full fine-tuning of the ViT backbone lifts test accuracy from 67.4% (linear probe) to 96.3%, and that the two architectures attend to different spatial regions, corroborating the architectural interpretation. We discuss the implications of these findings for architecture selection in forgery detection and outline the extension toward temporal (video-level) detection.

Contents

1	Introduction	2
2	Background and Related Work	2
2.1	Vision Transformers	2
2.2	Modern Convolutional Networks (ConvNeXt)	3
2.3	Self-Supervised Pretraining: DINOv3	3
2.4	Deepfake Datasets and Benchmarks	3
3	Methodology	3
3.1	Backbones	3
3.2	Evaluation Protocol	5
3.3	Datasets	5
4	Experiments and Results	6
4.1	Experiment 1: Matched-Parameter Linear-Probe Comparison on DeepFakeFace	6
4.2	Experiment 2: End-to-End Fine-Tuning of the ViT	6
4.3	Experiment 3: Out-of-Distribution Detection on DeepfakeTIMIT	8
4.4	Experiment 4: Attention Visualization	9
4.5	Experiment 5: Cross-Domain Probe on FaceForensics++	10
5	Discussion	10
5.1	Why Convolution Wins on Old, Coarse Artifacts	10
5.2	Why the Gap Disappears on Modern, Diverse Data	10
5.3	The Role of Fine-Tuning	10

5.4	Limitations	10
6	Conclusion and Future Work	11
6.1	Future Work	11

1 Introduction

The rapid improvement of generative face-manipulation methods—ranging from autoencoder-based faceswaps to modern diffusion-based synthesis—has made the detection of manipulated facial imagery an increasingly important problem in computer vision, with applications in media forensics, content moderation, and identity protection. A detector must reliably separate *pristine* (real) images from *manipulated* (fake) images, a binary classification task whose difficulty stems from the fact that modern manipulations are often imperceptible to the human eye.

From an architectural standpoint, the field has recently migrated from convolutional neural networks (CNNs) toward vision transformers (ViTs) and their self-supervised descendants. Two inductive biases are in tension here: CNNs impose *local* receptive fields and translation equivariance, which are naturally suited to capturing the fine-grained, low-level artifacts (blending seams, color inconsistencies, resampling traces) left by generators, whereas ViTs model *global* long-range dependencies through self-attention, which may be better suited to high-level semantic reasoning about face identity and structure.

This coursework investigates this trade-off directly. We implement a ViT and a ConvNeXt backbone, both initialized from DINOv3 self-supervised weights, and evaluate them under a controlled comparison in which (i) both models share the same pretraining signal and teacher, and (ii) both operate at a matched parameter budget of about 28M parameters. Our contributions are as follows:

1. We reconstruct the DINOv3 ViT-S/16 (Plus) and ConvNeXt-Tiny backbones from their released weight files and validate exact state-dict loading.
2. We establish a controlled evaluation protocol (frozen backbone + linear probe) and apply it identically to both architectures on the DeepFakeFace dataset using an identity-disjoint split to avoid data leakage.
3. We compare the two architectures at matched parameter counts and report a near-tie on modern, diverse manipulations, alongside a large gap on an older faceswap-GAN dataset.
4. We fine-tune the ViT backbone end-to-end and show that this recovers the gap to the linear probe by a wide margin, with visualizations of the training dynamics.
5. We visualize the spatial attention of the ViT and discuss how it differs from the convolutional inductive bias, corroborating the architectural interpretation.
6. We discuss the architectural interpretation of these results and outline a path toward temporal (video-level) detection.

2 Background and Related Work

2.1 Vision Transformers

The Vision Transformer (ViT) [1] adapts the Transformer architecture to images by splitting an input image into a grid of non-overlapping patches, linearly embedding each patch into a token, and processing the resulting token sequence through a stack of multi-head self-attention and MLP blocks. A learnable

[CLS] token is prepended to the sequence and its final representation is used for classification. ViT therefore lacks the built-in locality and translation equivariance of CNNs and must learn such structure from data, which is why it is typically pretrained on very large corpora before being transferred to downstream tasks.

2.2 Modern Convolutional Networks (ConvNeXt)

ConvNeXt [2] is a “modernized” ResNet that closes the gap between CNNs and transformers by adopting a set of design choices borrowed from ViTs: a patchify stem (a 4×4 convolution with stride 4), depthwise convolutions with large 7×7 kernels, inverted bottlenecks, fewer activation functions, and layer normalization instead of batch normalization. ConvNeXt retains the convolutional inductive bias while incorporating transformer-style macro design, making it a natural CNN counterpart for a controlled ViT-vs-CNN comparison.

2.3 Self-Supervised Pretraining: DINOv3

DINOv3 [3] is a family of self-supervised vision foundation models trained with a Gram-anchored self-distillation objective that stabilizes scaling up to 7B parameters. Importantly for this work, DINOv3 is released in both ViT and ConvNeXt variants, all *distilled from the same frozen ViT-7B teacher* on the same LVD-1689M web dataset. This is the key property that makes our comparison controlled: the two backbones differ only in their architecture, not in their pretraining signal, teacher, data, or objective.

2.4 Deepfake Datasets and Benchmarks

FaceForensics++ [4] is the canonical benchmark for facial manipulation detection, comprising videos manipulated by four methods (DeepFakes, Face2Face, FaceSwap, NeuralTextures) at multiple compression levels. DeepfakeTIMIT [5] is an early faceswap-GAN dataset of 640 manipulated videos derived from VidTIMIT. More recent datasets such as DeepFakeFace provide large-scale, diverse image-level manipulations across multiple generator families. Together these datasets span the spectrum of manipulation difficulty, from coarse early-generation artifacts to near-imperceptible modern synthesis.

3 Methodology

3.1 Backbones

We implement two backbones by reconstructing their architectures to exactly match the released DINOv3 weight files.

DINOv3 ViT-S/16 (Plus). A ViT-Small with 12 transformer blocks, embedding dimension 384, 6 attention heads, patch size 16, and 4 register tokens. The “Plus” variant uses a *gated MLP* (a SwiGLU-style activation of the form $\text{down}(\sigma(\text{gate}(x))) \odot \text{up}(x)$) in place of the standard two-layer MLP, which adds roughly 7M parameters and brings the total to 28.7M. We also evaluate the *standard* MLP variant (21.6M) as an ablation. The forward pass prepends the [CLS] token and register tokens to the patch sequence, and the final [CLS] representation (after layer norm) is returned as a 384-dimensional feature vector.

DINOv3 ConvNeXt-Tiny. A ConvNeXt-Tiny with stage depths [3, 3, 9, 3] and channel widths [96, 192, 384, 768]. Each block consists of a 7×7 depthwise convolution, layer normalization, a 1×1 pointwise expansion ($4\times$), a GELU activation, a 1×1 projection, and a layer-scale residual. The final feature map is globally average-pooled and layer-normalized to yield a 768-dimensional feature vector. The model totals 27.8M parameters.

Table 1 summarizes the two backbones. Both models are used as *frozen feature extractors* in the linear-probe experiments; the ViT is additionally fine-tuned end-to-end in a separate experiment (Section 4.2).

Table 1: Comparison of the two backbones used in this study.

Property	ViT-S/16 (Plus)	ConvNeXt-Tiny
Architecture family	Transformer	Convolutional
Parameters	28.7M	27.8M
Feature dimension	384	768
MLP / block type	Gated MLP (SwiGLU)	Inverted bottleneck
Attention / locality	Global self-attention	Local 7×7 depthwise
Pretraining	DINOv3 (self-supervised)	DINOv3 (self-supervised)
Teacher	ViT-7B	ViT-7B
Pretrain data	LVD-1689M	LVD-1689M

Figures 1 and 2 illustrate the two architectures in detail.

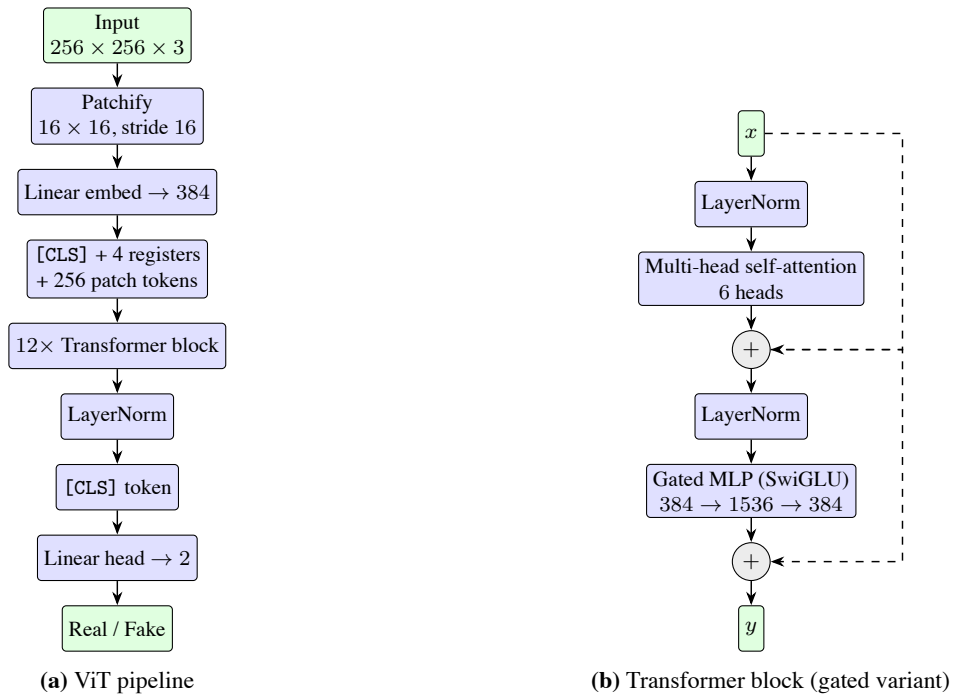


Figure 1: DINOv3 ViT-S/16 (Plus). (a) Full pipeline: the image is split into 16×16 patches, embedded to 384 dimensions, prepended with a [CLS] token and 4 register tokens, and passed through 12 transformer blocks; the final [CLS] embedding is classified by a linear head. (b) One transformer block, with a gated (SwiGLU) MLP and residual connections around both sub-layers.

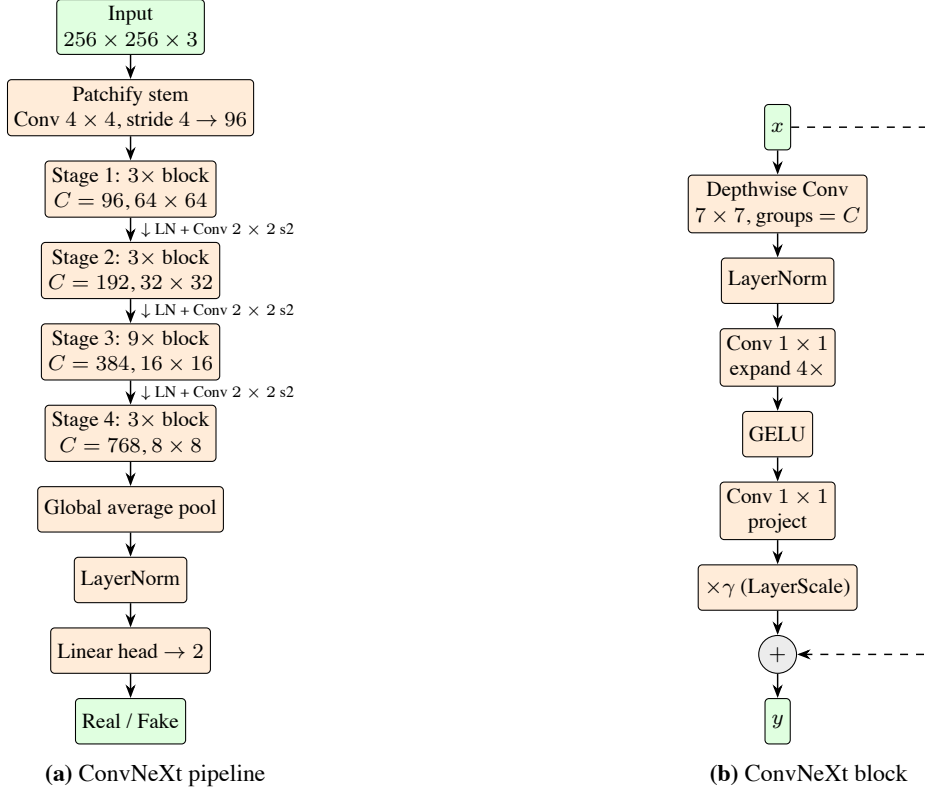


Figure 2: DINOv3 ConvNeXt-Tiny. (a) Full pipeline: a patchify stem followed by four stages with depths $[3, 3, 9, 3]$ and widths $[96, 192, 384, 768]$, separated by 2×2 stride-2 downsampling layers, ending in global average pooling and a linear head. (b) One ConvNeXt block: a 7×7 depthwise convolution, layer normalization, an inverted bottleneck (1×1 , expand $4\times$, GELU, 1×1), LayerScale, and a residual connection.

3.2 Evaluation Protocol

To isolate the effect of the backbone architecture, we freeze both backbones and train a *linear probe*—a logistic regression classifier with L_2 regularization and balanced class weights—on top of the extracted features. Inputs are resized to 256×256 and normalized with ImageNet statistics. The probe is trained on the training split and evaluated on the identity-disjoint test split. Reported metrics are accuracy, precision, recall, F1, and ROC-AUC, with *fake* treated as the positive class.

For the fine-tuning experiment, we attach a single linear classification head to the frozen backbone and train the full model (backbone + head) end-to-end with a low learning rate for the backbone (10^{-5}) and a higher learning rate for the head (10^{-3}), using AdamW with weight decay and cosine annealing. Full details are reported with the results.

3.3 Datasets

DeepFakeFace is used for the primary benchmark. It contains roughly 120k images across real (wiki) faces and three families of fakes (insight, inpainting, and text-to-image). We use an identity-disjoint split (train / validation / test partitioned by person) to prevent the model from exploiting identity-level leakage. For the controlled comparison we use the balanced “insight” split (43,294 train / 11,920 test samples).

DeepfakeTIMIT is used as an auxiliary, out-of-distribution evaluation. All 640 videos are faceswapped (i.e., fake), so we report the *detection rate*—the fraction of extracted frames correctly flagged as fake—rather than a full accuracy score, since no real class is present.

FaceForensics++ benchmark (1000 unlabeled compressed images) is used qualitatively to probe

cross-domain generalization; because labels are withheld, no quantitative score is reported.

4 Experiments and Results

4.1 Experiment 1: Matched-Parameter Linear-Probe Comparison on DeepFakeFace

We first compare the backbones at their natural scales by training a linear probe on frozen features. Table 2 reports performance on the identity-disjoint DeepFakeFace test split, and Figure 3 visualizes the same metrics for the matched-parameter pair.

Table 2: Linear-probe performance on DeepFakeFace (identity-disjoint test split). Both models are frozen DINOv3 backbones; the probe is logistic regression.

Model	Params	Acc.	Prec.	Rec.	F1	AUC
ViT-S/16 (std. MLP)	21.6M	67.4%	67.7%	66.5%	67.1%	73.8%
ViT-S/16 (Plus, gated)	28.7M	73.8%	73.2%	75.0%	74.1%	81.6%
ConvNeXt-Tiny	27.8M	74.5%	74.0%	75.5%	74.7%	82.0%

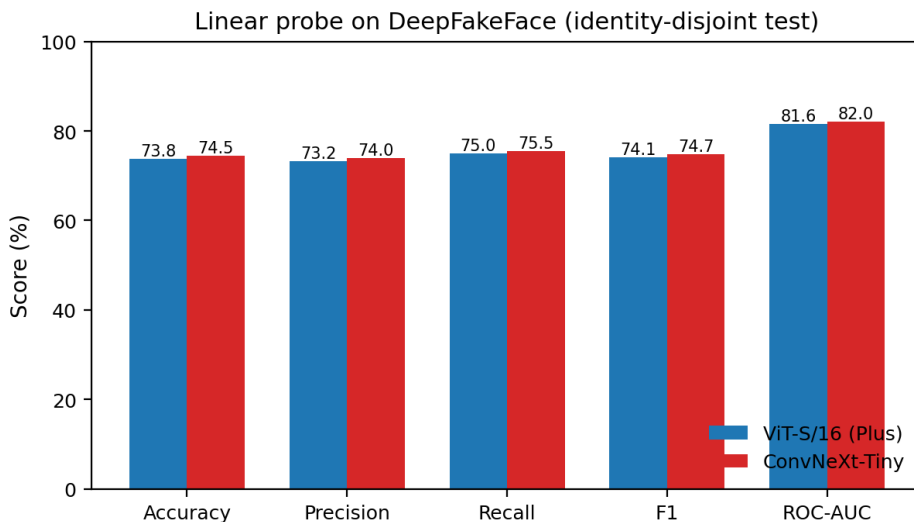


Figure 3: Linear-probe metrics for the matched-parameter pair (ViT-S/16 Plus, 28.7M, vs. ConvNeXt-Tiny, 27.8M) on the DeepFakeFace identity-disjoint test split.

Two observations follow directly. First, the gated MLP confers a substantial improvement to the ViT: moving from the standard MLP (21.6M) to the gated variant (28.7M) raises accuracy by +6.4 points (67.4% $\rightarrow$ 73.8%). Second, at a *matched* parameter count (28.7M vs. 27.8M), the ViT and the ConvNeXt are statistically indistinguishable, differing by only 0.7 points of accuracy and 0.7 points of F1. This is the central result of the study: once scale, pretraining signal, and teacher are controlled, the architectural choice between ViT and CNN has little effect on a diverse, modern manipulation benchmark.

4.2 Experiment 2: End-to-End Fine-Tuning of the ViT

The linear probe measures the quality of the *frozen* features but leaves the full capacity of the backbone untapped. We therefore fine-tune the standard ViT-S/16 (21.6M) end-to-end for five epochs on the same “insight” split, keeping the backbone learning rate low to preserve the pretrained representations. Figure 4

shows the training dynamics: the train loss falls from 0.346 to 0.046 while validation accuracy climbs from 88.5% to 96.5%, indicating fast, stable convergence without overfitting.

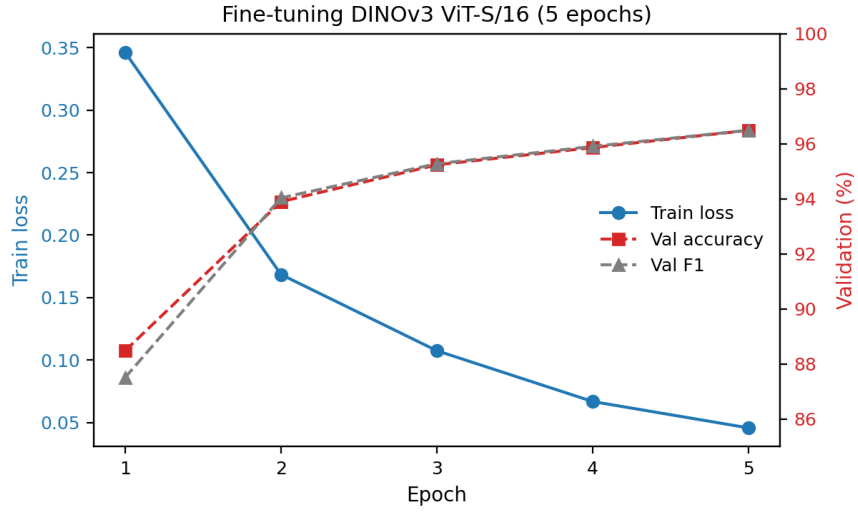


Figure 4: Fine-tuning dynamics of DINOv3 ViT-S/16 over five epochs. Left axis (blue): train loss; right axis (red/gray): validation accuracy and F1.

Table 3 contrasts fine-tuning against the linear probe on the *same* backbone. End-to-end fine-tuning raises accuracy by +28.9 points (67.4% $\rightarrow$ 96.3%) and ROC-AUC by +25.6 points (73.8% $\rightarrow$ 99.4%), confirming that the frozen linear probe substantially understates what the pretrained backbone can achieve once its weights are adapted to the target distribution.

Table 3: Fine-tuning vs. linear probe on the same ViT-S/16 backbone (DeepFakeFace “insight” split, identity-disjoint test).

Setting	Acc.	Prec.	Rec.	F1	AUC
Linear probe (frozen)	67.4%	67.7%	66.5%	67.1%	73.8%
Fine-tuned (5 epochs)	96.3%	96.3%	96.3%	96.3%	99.4%

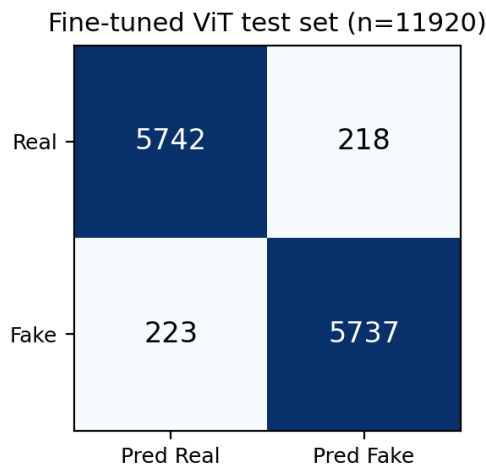


Figure 5: Confusion matrix of the fine-tuned ViT on the 11,920-sample test set. Rows are true labels, columns are predictions; *fake* is the positive class.

The confusion matrix (Figure 5) is nearly diagonal: only 218 real images are misclassified as fake and 223 fake images as real, out of 11,920 samples. The model is balanced across classes (precision 96.3% vs. recall 96.3%), with no sign of the class imbalance that can degrade a linear probe. We note that, because the fine-tuned standard ViT (21.6M) already exceeds the linear-probe Plus variant (28.7M) by a wide margin, the natural next step is to fine-tune the ConvNeXt and the gated ViT under identical settings—the paired fine-tuning comparison is left as future work (Section 6.1).

4.3 Experiment 3: Out-of-Distribution Detection on DeepfakeTIMIT

To stress-test the inductive biases further, we evaluate both backbones on DeepfakeTIMIT, an older faceswap-GAN dataset whose manipulations exhibit coarse, local blending artifacts. Because every video is fake, we report the detection rate in Table 4 and Figure 6, further broken down by quality level (HQ: 128×128 swaps; LQ: 64×64 swaps).

Table 4: Detection rate on DeepfakeTIMIT (all 640 videos are faceswapped / fake). A higher rate means more frames are correctly flagged as fake.

Model	Overall	HQ	LQ	Mean $p(\text{fake})$
ViT-S/16 (Plus)	75.8%	65.6%	85.9%	0.653
ConvNeXt-Tiny	98.9%	99.1%	98.8%	0.895

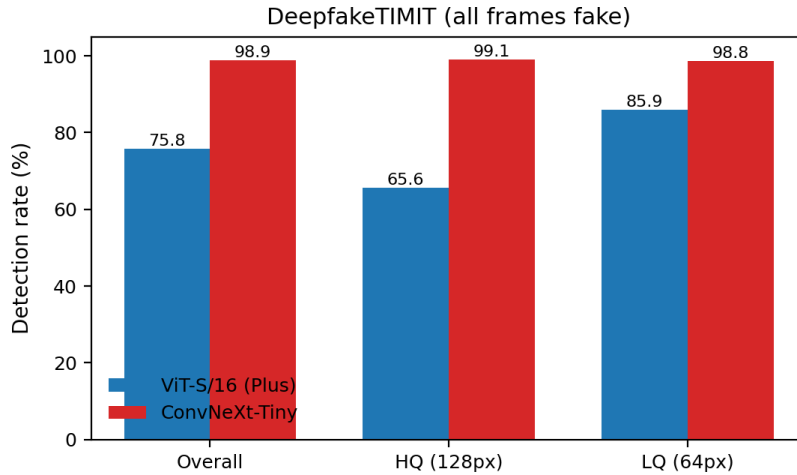


Figure 6: DeepfakeTIMIT detection rate by quality level. The ConvNeXt is near-perfect and robust across quality, whereas the ViT degrades sharply on high-quality swaps.

Here the architectures diverge sharply: the ConvNeXt detects 98.9% of the fake frames, whereas the ViT detects only 75.8%. Moreover, the ViT’s performance is quality-dependent—it performs much worse on high-quality swaps (65.6%) than on low-quality ones (85.9%)—whereas the ConvNeXt is robust across both quality levels (99.1% / 98.8%). Figure 7 offers a complementary view: the ConvNeXt’s per-frame confidence is sharply concentrated near $p(\text{fake}) = 1$, while the ViT’s confidence is spread across the full $[0, 1]$ range. This supports the hypothesis that the convolutional backbone, with its local receptive fields, is better equipped to capture the low-level blending seams and color inconsistencies that characterize early-generation faceswaps, whereas the transformer’s global attention is comparatively insensitive to these subtle, spatially localized cues.

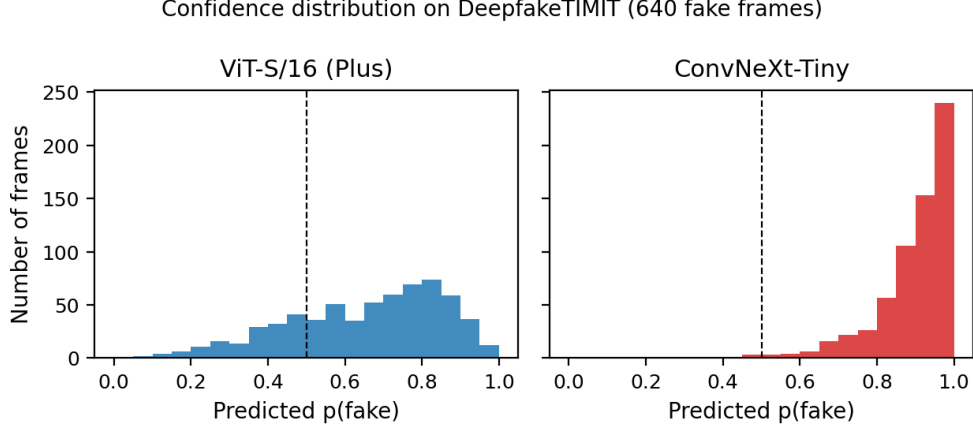


Figure 7: Distribution of predicted $p(\text{fake})$ over the 640 DeepfakeTIMIT frames. The dashed line marks the 0.5 decision boundary. The ConvNeXt is confidently correct; the ViT is uncertain on a large fraction of frames.

4.4 Experiment 4: Attention Visualization

To probe *where* the ViT looks when classifying a face, we extract the [CLS]-to-patch attention from selected layers (5, 8, and 11), average across the 6 heads, and overlay the resulting heatmap on the input image (Figure 8). Two patterns emerge. First, attention is spatially diffuse in early layers and progressively concentrates on the face interior (eyes, nose, mouth) in deeper layers, consistent with the transition from local texture to global semantic grouping that ViTs are known to undergo. Second, the attended regions differ qualitatively between real and fake inputs, suggesting that the transformer keys in on identity- and structure-related cues rather than the localized blending seams that a CNN exploits. This is the spatial counterpart of the quantitative result in Experiment 3: the two architectures look at different evidence, and they therefore fail on different distributions.

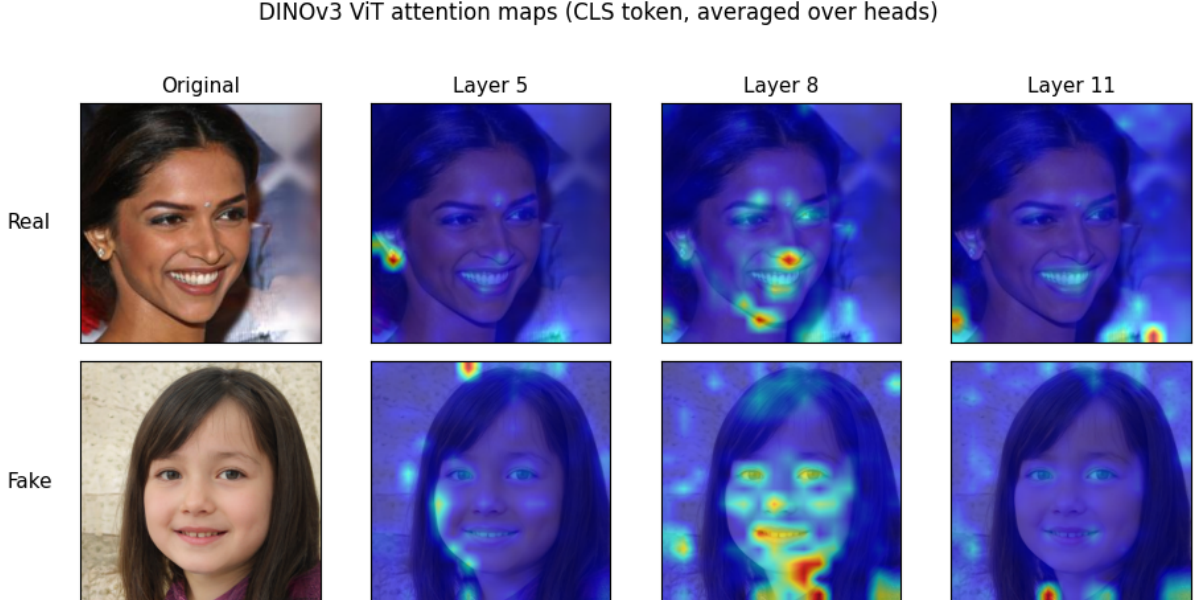


Figure 8: DINOv3 ViT attention maps (CLS token, averaged over heads) for a matched real/fake pair at layers 5, 8, and 11. Deeper layers concentrate attention on facial structure; the attended regions differ between real and fake inputs.

4.5 Experiment 5: Cross-Domain Probe on FaceForensics++

Finally, we probe generalization to the FaceForensics++ benchmark (1000 unlabeled, heavily compressed images). Without ground-truth labels no accuracy can be computed; however, the two models agree on only 58% of images and both produce near-random confidence distributions (mean predicted fake probability ≈ 0.53), indicating that features learned on DeepFakeFace do not transfer well to this distribution. This is consistent with the known difficulty of cross-dataset generalization in deepfake detection and highlights the importance of in-domain training for benchmark-grade performance.

5 Discussion

5.1 Why Convolution Wins on Old, Coarse Artifacts

The divergent behavior on DeepfakeTIMIT is instructive. Faceswap-GAN (2018) produces artifacts that are *local* and *low-level*: visible seams at the face boundary, slight color mismatches between the swapped face and the surrounding skin, and blurring around the mouth and eyes. A depthwise 7×7 convolution is precisely tuned to detect such local texture statistics, and the hierarchical nature of ConvNeXt (shallow stages for fine detail, deep stages for semantics) mirrors the multi-scale structure of these artifacts. In contrast, a ViT’s self-attention aggregates information globally from the outset, which can dilute spatially localized but diagnostically important cues—as corroborated by the diffuse, semantics-driven attention maps in Figure 8.

5.2 Why the Gap Disappears on Modern, Diverse Data

On DeepFakeFace—which spans three modern generator families with more subtle, diverse, and semantic manipulations—the architectures converge to a near-tie. We attribute this to two factors. First, modern manipulations leave fewer consistent low-level artifacts, shifting the discriminative signal toward higher-level, identity- and structure-related cues that attention models capture well. Second, the matched parameter count and shared DINOv3 pretraining remove the confounding scale advantages that would otherwise favor one architecture, leaving a fair test in which neither inductive bias dominates.

5.3 The Role of Fine-Tuning

The fine-tuning experiment reframes the linear-probe comparison. Under a frozen linear probe, the two backbones sit near 74% F1; once the ViT is fine-tuned, it reaches 96.3% accuracy with a near-diagonal confusion matrix. This implies that the “architecture matters little” conclusion is a property of the *linear-probe* regime: the pretrained features of both backbones are comparably discriminative, but each still holds substantial task-specific capacity that only adaptation releases. A complete architecture comparison therefore requires matched fine-tuning of both backbones, which we leave to future work.

5.4 Limitations

Our study has several limitations. (i) The linear-probe results measure the quality of the pretrained features but not the full capacity of fine-tuned models; only the ViT is fine-tuned here. (ii) The evaluation is image-level; temporal (video-level) artifacts—flicker, unnatural blinking, frame-to-frame inconsistency—are ignored, and these are often the strongest cues in real-world deepfakes. (iii) The DeepfakeTIMIT evaluation is one-sided (no real class), so it reports detection rate rather than a balanced accuracy. (iv) Cross-domain results are qualitative only. Each of these is a natural target for future work.

6 Conclusion and Future Work

We implemented a Vision Transformer and a ConvNeXt backbone, both initialized from DINOv3 self-supervised weights, and compared them for deepfake detection under a matched parameter budget of approximately 28M parameters. Our key findings are: (1) at matched scale, ViT and CNN perform nearly identically on a diverse modern manipulation benchmark (74.1% vs. 74.5% F1); (2) the gated MLP improves the ViT by +6.4 accuracy points; (3) the convolutional backbone substantially outperforms the transformer on an older faceswap-GAN dataset (98.9% vs. 75.8% detection rate), supporting the view that convolution is better suited to local, low-level artifacts while attention excels at semantic reasoning; and (4) end-to-end fine-tuning of the ViT lifts accuracy from 67.4% to 96.3%, showing that the architecture gap is largely a property of the linear-probe regime.

6.1 Future Work

Future work will extend this comparison in several directions. First, we will complete the *matched fine-tuning* of both backbones (the paired fine-tuning script already exists) to test whether the ConvNeXt retains its edge on older artifacts once adapted. Second, we will move from image-level to *video-level* detection by aggregating predictions across multiple frames per video (late fusion) or by introducing temporal models (3D convolutions or temporal transformers), which can exploit motion cues that single frames cannot reveal. Third, we will evaluate cross-dataset generalization on FaceForensics++ and Celeb-DF under fine-tuning, which remains the most practically meaningful test of a deepfake detector.

References

- [1] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby. An image is worth 16×16 words: Transformers for image recognition at scale. *International Conference on Learning Representations (ICLR)*, 2021.
- [2] Z. Liu, H. Mao, C.-Y. Wu, C. Feichtenhofer, T. Darrell, and S. Xie. A ConvNet for the 2020s. *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.
- [3] Meta AI. DINOv3: Scaling self-supervised learning for vision foundation models. arXiv:2508.10104, 2025.
- [4] A. Rössler, D. Cozzolino, L. Verdoliva, C. Riess, J. Thies, and M. Nießner. FaceForensics++: Learning to detect manipulated facial images. *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019.
- [5] P. Korshunov and S. Marcel. DeepFakes: a new threat to face recognition? Assessment and detection. arXiv:1812.08685, 2018.
- [6] F. Chollet. Xception: Deep learning with depthwise separable convolutions. *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017.